N L Subramanya Tejas

nlsubramanyatejas@gmail.com linkedin.com/in/tejasnls tejasnls.dev

Experience

American Express (via Cognizant) Dec 2020 – Present
Engineer II *Bengaluru, India*

Enterprise Search Platform

Architecture & Platform Engineering

- Led a 6-month architectural migration extracting Global Search into a dedicated OneApp MFE, establishing independent release cycles for a platform processing ∼2.7M global searches monthly (1.9M US, 808k International).
- Designed an Akamai rewrite strategy to seamlessly serve the new Search application behind the existing user-facing URL, preserving the user experience without browser-level redirects.
- Implemented 200ms debouncing and in-flight request cancellation for typeahead APIs, drastically reducing unnecessary network traffic and preventing race conditions across 2.75M monthly interactions.
- Architected the frontend foundation for a new internal Search as a Service workflow platform, designing a scalable root-and-child module architecture with granular RBAC and SSO.

AI & Search Initiatives

- Led frontend delivery of intent-aware semantic typeahead powered by hybrid retrieval, driving a 4% lift in Click-Through Rate (CTR) and a 6% improvement in search serve rate.
- Designed deterministic GUID-based test and control bucketing for phased rollouts, safely managing the 50% canonical release of semantic typeahead in the US market to serve 2.75M monthly requests.
- Built a first-party suggestion analytics pipeline capturing end-to-end user telemetry, eliminating a 10-second tracking blind spot and establishing KPI measurement via enterprise data pipelines.

Leadership & Engineering Excellence

- Modernized frontend practices across UI modules by establishing a release-oriented OneFlow branching strategy, refactoring legacy class-based components, and removing code smells.

Partner Servicing Portal – Enterprise Onboarding Platform

- Reduced page load time by 30% through code-splitting and performance profiling, and improved accessibility scores by 45% to meet WCAG standards.
- Reduced reported security vulnerabilities by 60% through secure coding practices and PRISMA scanning; migrated the test stack from Enzyme to React Testing Library.

Projects

ForexFix – Travel FX Intelligence Platform | *Next.js 15*, *TypeScript*, *Supabase*, *Gemini LLM*, *Tailwind CSS*

- Architected a hybrid system that isolates deterministic financial calculations from Gemini LLM-generated insights, ensuring strict numerical accuracy while leveraging AI for contextual travel recommendations.
- Leveraged AI-assisted development workflows to accelerate implementation, deploying a production-ready Vercel application achieving near-perfect Lighthouse scores (98 Performance, 96 Accessibility, 100 SEO).

Technical Skills

Frontend Engineering: JavaScript (ES6+), TypeScript, React, Next.js, Tailwind CSS, Micro-Frontend Architecture, Redux, Context API, Performance Optimization, Accessibility (WCAG)

Backend & Cloud: Node.js, REST APIs, API Integration, Authentication (SSO, JWT), Supabase, Vercel

Testing & Quality: Jest, React Testing Library, Cucumber (BDD), Unit & Integration Testing, SonarQube, PRISMA

AI & Search: LLM Application Development, Prompt Engineering, Context Engineering, RAG, Semantic/Hybrid Search, Embeddings, LangChain, LangSmith, Flowise

DevOps & Observability: CI/CD (GitHub Actions, Jenkins), Docker, Akamai, Splunk, OpenSearch, Adobe Analytics

Education

Atria Institute of Technology (VTU) 2016 – 2020
B.E. in Computer Science & Engineering *Bengaluru, India*